

Speaker Verification for Sinhala and Tamil: A Reproducible Benchmark with Zero-Shot and Training-Free Adaptation Baselines

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Abstract—Sinhala and Tamil, the two official languages of Sri Lanka, have no published speaker-verification (SV) benchmark. We present the first reproducible SV benchmark for these languages, built from public OpenSLR read-speech corpora: 527 speakers (478 Sinhala, 49 Tamil) with 3,000 target and 9,000 impostor trials per language, deterministically generated from a single seed. On this benchmark we report the first zero-shot baselines for four state-of-the-art VoxCeleb-trained systems (ECAPA-TDNN and ReDimNet-B1/B2/B6): equal-error rates of 2.71–4.78% on Sinhala and 1.48–3.21% on Tamil, i.e., 2–8× their published VoxCeleb1-O figures even on clean read speech, with poor minimum detection cost (0.24–0.48 at $P_{tgt}=0.01$) throughout. A training-free adaptation study shows that adaptive score normalization with a small in-domain cohort recovers up to 39% relative EER (best system: 2.07%/0.90%), that PLDA helps or hurts depending on corpus recording structure, and that swapping calibrators between the two languages inflates C_{lr} by up to 5.7×, making per-language calibration mandatory. In-language fine-tuning of WavLM confirms and completes this picture. Trial lists, code, and configurations are released.

Index Terms—Speaker verification, low-resource languages, benchmark, score normalization, calibration.

I. INTRODUCTION

SPEAKER verification (SV) on English benchmarks has matured: publicly released systems reach 0.4–0.8% equal-error rate (EER) on VoxCeleb1-O [1]–[3]. How well these released checkpoints transfer to unseen low-resource languages is far less well characterized. For the two official languages of Sri Lanka — Sinhala (Indo-Aryan, ~17 million speakers) and Tamil (Dravidian, ~5 million Sri Lankan speakers) — the gap is complete: neither language appears in any catalogued SV corpus, and we could find no published SV results for modern architectures on Sinhala at all.

Evidence from other languages suggests transfer is costly: VoxCeleb-trained systems degrade by a factor of 2–4 in EER on CN-Celeb [4], and cross-lingual trials suffer systematic score shifts that break calibration [5], [6]. Whether these findings extend to Sinhala and Tamil, whether the published VoxCeleb ranking of systems is preserved under the shift, and how much of the loss is recoverable *without any training* are open questions with direct deployment relevance.

This letter contributes:

- **SL-SPV benchmark v0** — the first reproducible SV benchmark for Sinhala and Tamil: 527 speakers drawn from public OpenSLR corpora [7], [8], 12,000 trials per language constructed deterministically from a single seed, and evaluation at two operating points (minDCF at $P_{tgt} = 0.01$ and 0.05) alongside EER.
- **Zero-shot baselines** for four state-of-the-art released checkpoints — ECAPA-TDNN [1], [9] and ReDimNet-B1/B2/B6 [2] — quantifying the English→Sinhala/Tamil degradation at 2–8× published VoxCeleb1-O EER.
- **A training-free adaptation study** of adaptive score normalization (AS-Norm) [5], two-covariance PLDA [10], [11], and score calibration, including the first cross-lingual calibration-transfer measurement for this language pair. We additionally verify the conclusions against in-language fine-tuning of WavLM [12].

Section II describes the benchmark, Section III the zero-shot results, Section IV the adaptation study, Section V the limitations, and Section VI concludes.

II. THE SL-SPV BENCHMARK (V0)

A. Corpora

Sinhala audio comes from OpenSLR SLR52 [7], a crowd-sourced read-speech corpus recorded at 16 kHz; retaining speakers with at least 10 utterances yields **478 speakers** (~185k utterances). SLR52 publishes no gender or session metadata. Tamil audio comes from OpenSLR SLR65 [8], a studio-recorded read-speech corpus (48 kHz, downsampled to 16 kHz), yielding **49 speakers** (4,286 utterances) with speaker gender recoverable from the file naming convention. Both corpora are CC BY-SA licensed, so the benchmark is fully redistributable as trial lists over public audio.

B. Trial protocol

Each speaker’s utterances are split 80/20 into a training/adaptation pool and a held-out test pool (seed 42); all trials are drawn from test utterances only. Per language we sample **3,000 target and 9,000 impostor trials** (1:3). Target trials pair same-speaker, same-language utterances. Impostor trials are same-language and *same-gender whenever gender is known for both speakers*: the Tamil list contains zero cross-gender impostor pairs, while Sinhala impostors are not gender-controlled because SLR52 lacks the metadata (Section V). Pairs are deduplicated as unordered pairs, self-pairs are excluded, and all sampling is reproducible from the single seed.

Manuscript received [Month] [Day], [Year]; revised [Month] [Day], [Year].

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Code, configurations, and trial lists are released at [github.com/\[user\]/voxceleb_trainer-SL-SPV](https://github.com/[user]/voxceleb_trainer-SL-SPV).

TABLE I

ZERO-SHOT RESULTS ON SL-SPV v0 (FULL 12,000-TRIAL LISTS PER LANGUAGE): EER (%) AND minDCF AT $P_{tgt} = 0.01/0.05$. “V1-O” IS EACH CHECKPOINT’S PUBLISHED VOXCELEB1-O EER (%).

System	Par.	V1-O	Sinhala		Tamil	
			EER	DCF _{.01/.05}	EER	DCF _{.01/.05}
ECAPA-TDNN [1]	20M	0.80	4.78	.471/.291	3.21	.482/.207
ReDimNet-B1 [2]	2.2M	0.73	3.57	.306/.209	1.67	.406/.162
ReDimNet-B2 [2]	4.7M	0.52	4.17	.422/.244	1.97	.454/.170
ReDimNet-B6 [2]	15M	0.37	2.71	.238/.153	1.48	.422/.160

C. Metrics

We report EER (mean of false-positive and false-negative rates at the crossing) and minDCF at $P_{tgt} \in \{0.01, 0.05\}$ with $C_{miss}=C_{fa}=1$, making the numbers comparable with both the NIST and VoxCeleb literatures. Calibration experiments additionally report C_{llr} . The evaluation harness treats unlabeled trials as a hard error rather than imputing labels.

III. ZERO-SHOT BASELINES

We evaluate four released VoxCeleb-trained checkpoints without any adaptation: the SpeechBrain ECAPA-TDNN [1], [9] trained on VoxCeleb1+2 [3], [13], and the large-margin fine-tuned ReDimNet-B1/B2/B6 checkpoints [2]. Scoring is cosine similarity on ℓ_2 -normalized full-utterance embeddings, with no score normalization or calibration — the raw transfer reference that Section IV improves on. Table I reports the results, the first published SV baselines for these languages. Four observations:

(i) *Language shift alone costs a multiple, not a margin.* Relative to each checkpoint’s published VoxCeleb1-O EER, the SL-SPV degradation factor spans $4.9\text{--}8.0\times$ (Sinhala) and $2.3\text{--}4.0\times$ (Tamil): Tamil sits within the $2\text{--}4\times$ band that the CN-Celeb literature [4] reports for in-the-wild Chinese, and Sinhala exceeds it, despite v0 being clean, single-session read speech.

(ii) *The VoxCeleb ranking mostly, but not fully, transfers.* ReDimNet-B6 is best and ECAPA-TDNN worst on both languages, yet B1 beats B2 on both languages despite B2’s better VoxCeleb1-O figure — and parameter count does not buy robustness monotonically: B1 (2.2M) outperforms both B2 (4.7M) and ECAPA-TDNN (~ 20 M).

(iii) *EER alone overstates how solved Tamil is.* Tamil EER is uniformly lower (studio audio, 49 speakers), but Tamil minDCF_{.01} is uniformly worse than Sinhala for the ReDimNets (B6: 0.422 vs. 0.238): with few speakers, a handful of hard impostors dominates the low-false-alarm operating point.

(iv) *Scores are usable but badly scaled.* minDCF_{.01} is poor everywhere (0.24–0.48) while EERs look deployable — the classic signature of uncalibrated scores in a mismatched domain [5], [6], and the direct motivation for the next section.

TABLE II

TRAINING-FREE BACKEND ADAPTATION ON SL-SPV v0: EER % (minDCF AT $P_{tgt}=0.01$). BOLD: BEST EER PER SYSTEM AND LANGUAGE.

System	Backend	Sinhala	Tamil
ECAPA-TDNN	cosine	4.78 (.471)	3.21 (.482)
	+AS-Norm	3.19 (.330)	2.24 (.480)
	PLDA	6.00 (.598)	2.46 (.495)
	+AS-Norm	5.80 (.482)	2.06 (.374)
ReDimNet-B1	cosine	3.57 (.306)	1.67 (.406)
	+AS-Norm	2.38 (.245)	1.47 (.433)
	PLDA	5.34 (.452)	1.77 (.419)
	+AS-Norm	5.23 (.382)	1.49 (.297)
ReDimNet-B6	cosine	2.71 (.238)	1.48 (.422)
	+AS-Norm	2.07 (.167)	0.90 (.376)
	PLDA	3.84 (.293)	1.11 (.340)
	+AS-Norm	3.56 (.254)	1.16 (.314)

IV. TRAINING-FREE ADAPTATION

A. Setup

All levers in this section require no fine-tuning — only one embedding-extraction pass over *training-pool* audio, which is disjoint from every trial utterance. From the training pool we build (a) a multi-language cohort of 527 utterances (one per speaker, both languages, following [5]) and (b) a PLDA training set of 10,480 utterances (≤ 20 per speaker, 527 speakers). Four backends are compared per model and language: (1) *cosine* (the Table I baseline, reproduced exactly as a sanity check); (2) *AS-Norm*: adaptive symmetric score normalization using, per trial side, the mean/std of its top- K ($K=300$) cohort similarities, $s' = \frac{1}{2}[(s - \mu_e)/\sigma_e + (s - \mu_t)/\sigma_t]$; (3) *PLDA*: two-covariance PLDA [10], [11] with LDA to 150 dimensions, centering and length normalization; and (4) PLDA followed by AS-Norm. We apply the grid to ECAPA-TDNN, ReDimNet-B1, and ReDimNet-B6 — the worst, smallest, and best zero-shot systems.

For calibration, each trial list is split 50/50 (stratified, seed 42) into calibration and evaluation halves; an affine logistic-regression mapping (score $\rightarrow$ log-likelihood ratio) is fit per condition. *Self-cal* fits and evaluates within one language; *cross-cal* fits on one language and evaluates on the other, measuring the cross-lingual score shift [6].

B. Results

1) *AS-Norm is the single most cost-effective lever:* Table II shows AS-Norm with the small in-domain cohort delivers 23–39% relative EER reduction in five of six model–language pairs (12% for B1 on Tamil) — bracketing the 20–30% band reported for multilingual score normalization [5] — and improves minDCF_{.01} in five of six pairs as well (best: B6 Sinhala 0.238 $\rightarrow$ 0.167; the exception is B1 Tamil, 0.406 $\rightarrow$ 0.433). The best training-free system is **ReDimNet-B6 + AS-Norm: 2.07% (Sinhala) / 0.90% (Tamil)**, at the cost of one cohort file and no training.

TABLE III
CROSS-LINGUAL SCORE SHIFT (REDIMNET-B6): C_{1lr} ON THE EVALUATION
HALF WITH SELF-LANGUAGE VS. CROSS-LANGUAGE CALIBRATION
(INFLATION FACTOR IN PARENTHESES).

Backend	Sinhala eval		Tamil eval	
	self	Tamil-fit	self	Sinhala-fit
cosine	0.129	0.489 (3.8 $\times$)	0.095	0.149 (1.6 $\times$)
+AS-Norm	0.099	0.565 (5.7 $\times$)	0.071	0.167 (2.4 $\times$)
PLDA	0.171	0.390 (2.3 $\times$)	0.073	0.084 (1.2 $\times$)
+AS-Norm	0.163	0.442 (2.7 $\times$)	0.066	0.094 (1.4 $\times$)

2) *PLDA is condition-dependent*: In-domain PLDA *hurts* Sinhala EER for every model (B6: 2.71 $\rightarrow$ 3.84) yet *helps* Tamil EER for two of three models (ECAPA: 3.21 $\rightarrow$ 2.46; B6: 1.48 $\rightarrow$ 1.11; B1: 1.67 $\rightarrow$ 1.77), and PLDA+AS-Norm gives the best Tamil minDCF_{.01} for all three models. Our working hypothesis is a benchmark artifact: the PLDA fit is dominated by the 478 Sinhala speakers of near-single-session crowdsourced audio, so its within-speaker covariance underestimates real channel variability, whereas the studio-recorded Tamil fits the model better. The claim that PLDA wins under domain shift [14] should therefore not be dismissed on this evidence; a multi-session corpus (Section V) is the decisive test.

3) *Cross-lingual score shift is large and real*: Table III reports C_{1lr} for ReDimNet-B6 under self- and cross-language calibration. Swapping calibrators across languages inflates C_{1lr} by up to 5.7 $\times$ (Sinhala scored with a Tamil-fit calibrator under AS-Norm): *per-language calibration is mandatory* for deployment, extending the cross-lingual score-shift finding of [6] to this language pair. Notably, PLDA scores are the most *transportable* across languages (cross-cal penalty only 1.2–2.7 $\times$) even where PLDA’s EER is worse — generatively modeled scores appear to live in a more language-invariant domain than cosine scores.

C. Verification against in-language fine-tuning

To check that the training-free conclusions are not an artifact of frozen encoders, we additionally fine-tuned WavLM-base+ [12] (attentive-statistics pooling, 192-d embeddings, AAM-softmax [15], margin 0.2, scale 30) on 31,049 training-pool utterances (all 527 speakers, capped at 60 utterances per speaker). Full fine-tuning reaches $1.69 \pm 0.06\%$ (Sinhala) and $1.41 \pm 0.10\%$ (Tamil) EER over three seeds — past the best zero-shot system — whereas LoRA [16] ($r=8$, 1.1% of parameters) reaches only $3.73 \pm 0.49\%$ / $3.11 \pm 0.61\%$ on this closed-set benchmark. Composing the backends with the fine-tuned encoder completes the domain-shift picture: PLDA becomes redundant or harmful on *both* languages (Sinhala 1.63 $\rightarrow$ 2.46 at seed 42), exactly as predicted for in-domain embeddings [14], while AS-Norm’s EER gain saturates but it still improves the operating-point metrics (minDCF_{.01} Sinhala 0.180 $\rightarrow$ 0.144, Tamil 0.175 $\rightarrow$ 0.149, with C_{1lr} down $\sim 15\%$). Backend adaptation thus remains worth keeping for deployment even once the encoder is adapted.

V. LIMITATIONS

The v0 benchmark has known limitations that any use of its numbers must respect. (1) Both corpora are read speech with, in effect, a single recording session per speaker, so absolute EERs are optimistic; the numbers are valid for *relative* comparison of systems and backends, not for comparison against in-the-wild benchmarks. (2) The evaluation is closed-set: the adaptation/training pool contains the same 527 speakers as the trials (utterance-disjoint), which is standard for domain-adaptation PLDA but must be stated — it also removes the overfitting penalty that usually favors parameter-efficient over full fine-tuning, so the LoRA result of Section IV-C should not be read as general. (3) Sinhala impostor trials are not gender-controlled (SLR52 has no gender metadata); Tamil trials are. (4) SLR65 Tamil is Indian-accented rather than Sri Lankan Tamil. (5) There is no telephony/codec condition and no code-switched speech. A v1 rerun on SLCeleb, a 280-speaker in-the-wild Sri Lankan corpus with multi-session and bilingual speakers (IEEE DataPort), is planned to address (1)–(4) and to provide open-set and genuine cross-lingual trials.

VI. CONCLUSION

This letter establishes the first reproducible speaker-verification benchmark for Sinhala and Tamil, publishes the first zero-shot baselines for four state-of-the-art VoxCeleb-trained systems on it, and characterizes what training-free adaptation can and cannot repair: AS-Norm with a 527-utterance cohort recovers up to 39% relative EER (best system 2.07%/0.90%), PLDA’s value hinges on corpus recording structure, and calibration does not transfer across the two languages. Trial lists, code, and configurations are released to make every number in this letter reproducible from a single seed and to serve as the reference point for the planned in-the-wild v1 benchmark.

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